Kaustubh Pandey

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Education

Manipal Institute of Technology — **2023 – Present**
B.Tech in Electronics & Communication (VLSI) — *Manipal, India*
- Current CGPA: 7.73 / 10 (as of 6th Semester)

Delhi Public School — **Udaipur, India**
CBSE – Secondary & Higher Secondary Education
- Class XII: 84.4% | Class X: 88.16%

Relevant Coursework

- VLSI Design
- Digital IC Design
- Analog IC Design
- Physical Design
- Analog Circuits
- FPGA System Design
- Semiconductor Physics
- Verilog HDL

Experience

VLSI System Design – PCB Design Intern — **Feb 2025 – Oct 2025**
PCB Designer — *Manipal, India*
- Designed and laid out the evaluation board for the Thejas32 (by Vega Processor & CDAC) using KiCad, including schematic capture and PCB layout.
- Collaborated on creating a shield for their FPGA board PCB design, integrating interface components and ensuring signal integrity.

Project MANAS – Hardware Subsystem, Autonomous UGV Team — **June 2024 – Present**
Hardware Engineer — *Manipal, India*
- Designed power distribution boards with high-current switching and implemented current sensing using current-sense ICs.
- Led PCB design (KiCad) for motor/microcontroller interfaces; integrated CAN-bus transceivers.
- **3rd place AutoNav Challenge (IGVC 2025)** and **PIMA Winner (ISDC 2025)** with autonomous UGV & UAV platforms.

Projects

AXI4-Stream Configurable FIR Filter IP Core | *SystemVerilog, Verification, FPGA* — **July 2026 – Present**
- Designing a parameterizable FIR filter IP core with an AXI4-Stream data path and AXI4-Lite control interface for runtime-reconfigurable coefficients.

Low-Dropout Regulator (LDO) Design | *Cadence Virtuoso, Calibre, Analog IC Design* — **Feb 2026 – June 2026**
- Designed a 180nm CMOS LDO in Cadence Virtuoso regulating Vout to 1.2V, using an OTA-based error amplifier with resistive feedback and a bandgap reference.
- Completed full-custom layout with DRC/LVS verification in Calibre; validated closed-loop DC regulation via simulation.

FPGA Lane & Obstacle Detection | *SystemVerilog, Vivado* — **Feb 2026 – June 2026**
- Designed a 6-stage SystemVerilog RTL pipeline (grayscale conversion, Gaussian blur, edge detection, ROI filtering, Hough transform) for real-time lane detection on an Artix-7 FPGA

PID Motor Controller | *Verilog, Vivado, Icarus Verilog, GTKWave* — **May 2025 – Nov 2025**
- **Published in IEEE:** DOI: 11232594 - Digital PID Controller for DC Motor Speed Regulation on FPGA.
- Designed Verilog PID controller with encoder feedback; synthesized pipelined architecture on Vivado FPGA for deterministic loops.

Technical Skills

Languages: SystemVerilog, Verilog, C++
Tools: Cadence Virtuoso, Calibre, Vivado, KiCad, Icarus Verilog, GTKWave, Synopsys Sentaurus TCAD, LTSpice
Technologies: PCB Design, FPGA, AMBA AXI4, Semiconductor Device Physics

Leadership / Extracurricular

IRC & ISDC 2026 (SPROS WEEK) Core Committee — **Jan 2026 – Feb 2026**
Core Committee Member — *Manipal Institute of Technology*
- Organized IRC and ISDC 2026 events as part of SPROS WEEK core committee.
- Coordinating technical competitions, workshops, and student participation.